Alex Dils

Berkeley, CA | dils@berkeley.edu | alex-dils.com | Scholar | LinkedIn | GitHub

Education

University of California, Berkeley — Expected May 2028
B.A. in ***Computer Science*** | GPA: 3.7/4.0
Coursework: Data Structures & Algorithms, Machine Structures, Discrete Mathematics & Probability, Multivariable Calculus
Activities: Machine Learning at Berkeley (ML@B), Internal & Research Committees

Technical Skills

Languages: Python, TypeScript, Java, C, SQL; **ML:** PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, SimpleITK; **Web/Data:** FastAPI, Node.js, React, PostgreSQL, Supabase, PostGIS; **Tools:** Docker, Slurm, Git

Experience

Founder & Engineer, Infrastruct — *Agentic workflow automation* — May 2026 – Present
- Built an approval-gated **AI workflow-automation platform** with a desktop observer and Node.js backend; implemented **49 workflow rules** across **15 app connectors**.
- Shipped traceable execution, rollback and idempotency safeguards, OAuth/SSO, billing, and deployment; sustained **176 ms median p95** across **2,500** production health requests at 100-way concurrency with zero failures.

Stanford Canary CREST Research Scholar, Stanford Canary Center / Gevaert Lab — Jun 2026 – Present
- Built a **cNF volume-estimation pipeline** for NF1 cancer-risk surveillance, combining center-coordinate lesion localization, SAM v2 masks, Depth Pro surface reconstruction, and per-lesion cm^3 integration.
- Generated synthetic RGB/depth lesions on Human Skin Repository 3D surfaces using physics-based protrusions and diffusion inpainting; fine-tuned Depth Pro, reducing depth-estimation AbsRel by **52.6%** and reaching $R^2 = 0.97$.

Machine Learning Engineer, Logitech — Feb 2026 – Jun 2026
- Built a **VLM evaluation harness** for meeting-room readiness and participant classification on Logitech's Rally Board 65, measuring task accuracy and inference latency across 55 models.
- Ran model and prompt ablations and identified Qwen3.5 4B as the leading deployment candidate for the target hardware and task.

Co-Founder, ML Engineering, Appraise AI — *AI fashion appraisal* — Nov 2025 – Mar 2026
- Led the ML roadmap and architecture for a resale-intelligence startup that raised **$15K** pre-seed.
- Built a **multimodal pricing engine** with **>95% top-3 accuracy** and Scrapy/Playwright ingestion collecting **500K+ listings/day**.

Research Intern (part-time), Stanford Center for Biomedical Informatics Research — Jun 2022 – Jun 2026
- Developed domain-robust skin-lesion segmentation methods and finite-element lung-motion augmentations that **improved SoTA tumor-segmentation Dice by >0.05**.
- Drafted sections of a Digital Twin manuscript and its journal submission cover letter; the research was **featured on Stanford Medicine's homepage**.

Projects

LanternTrace Explorer | alex-dils.com/lanterntrace — 2026
- Built a physics-based explorer to track and forecast spotted lanternfly spread across the U.S. East Coast.

Openleaf | alex-dils.com/openleaf — 2026
- Built a local macOS research workspace unifying LaTeX, PDF, Python, GitHub, SSH, and coding agents.

BsCode | alex-dils.com/bscode — 2026
- Built a desktop for coordinating Codex, Claude, and shell agents across local and remote workspaces.

Set Player | alex-dils.com/set-player — 2026
- Built a DJ-set memory layer linking recordings to Rekordbox history, timed tracks, lyrics, photos, and places.

Selected Publications & Manuscripts

** indicates first authorship.*

- *Enhancing Medical AI with Physiologically-Informed Data Augmentation*. Submitted to NeurIPS 2026.
  Contributing author: **Alex Dils**.
- **RGB–Depth Generation for cNF*. MICCAI MWM 2026.
  Christoph Sadée, **Alex Dils**, Krish Sangani, Lillian Rubino, Alexander Ferenchick, Varun Wadhwa, et al.
- **Eye For An Eye: A Deep-Learning and Analytical Method to Spatializing Stereoscopic Images*. National High School Journal of Science, 2025.
  Jake Yoshinaka, **Alex Dils**, and Leo McDonnell.
- **Addressing Bias and Confounders in AI-based Image Diagnosis — A Study of 117,610 Skin Lesions*. Manuscript under revision at *Cell*, 2025.
  Christoph Sadée, Serena Zhang, **Alex Dils**, Jared Bitz, Stefano Testa, Zhuo Ran Cai, et al.
- *Integrating Mechanistic Knowledge into Deep Learning for Improved Cancer Detection*. FEniCS Conference Proceedings, 2024.
  Christoph Sadée, Katherine Hartmann, **Alex Dils**, Ianto Xi, Francisco Carrillo-Perez, et al.